# Neil Janwani

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## EDUCATION

PH.D. **Robotics**, Georgia Institute of Technology — AUG 2024 – JUN 2029
Principal Advisor: Prof. Maegan Tucker (Dynamic Mobility Lab). — GPA: 4.0

B.S. **Computer Science**, California Institute of Technology — SEP 2020 – JUN 2024
Research Advisors: Prof. Aaron Ames, Prof. Joel Burdick. — GPA: 4.0

## RESEARCH SPOTLIGHT

**Multi-Objective Robotic Optimization.** JUN 2025 - Present
*Generating high-dimensional Pareto frontiers enables diverse personalization and tuning of learned controllers post-training.*

- Created GPU-native Multi-Objective RL Python toolbox [1] achieving up to 270× speedups over prior work.
- Leveraged hypernetworks to tractably approximate high-dimensional Pareto sets using multi-objective PPO.
- Generated diverse, high-quality whole-body locomotion across six objectives in ∼2 hours for the BRUCE humanoid.

**Robust and Stylized Humanoid Locomotion** JAN 2025 - Present
*Humanoid controllers that know when and how to deviate from idealized reference motions improves robustness.*

- Built a continuous gait library by smoothly interpolating between dynamically feasible reference motion pairs [2].
- Applied RL to robustly traverse the gait library, enabling higher imitation accuracy and finer control of gait style [2].
- Implemented adaptive framework for varying the trade-off between imitation-fidelity and robustness for humanoids.
- Achieved hardware deployment of aforementioned algorithms on the G1 Humanoid and BRUCE humanoid robots.

**Human-Robot Interaction and Personalization.** JUN 2021 - Present
*Robotic personalization and aligned human-robot teams can improve robot integration in the real world.*

- Devised quality-diverse exploration stage to familiarize users with their options before eliciting their preferences.
- Showed that initial exploration increases preference alignment on a synthetic oracle with non-stationary preferences.
- Improved safety-critical operator-robot alignment without losing guarantees with an intention-aware CBF [3].
- Designed motor control and gait-phase inference of a soft ankle exoskeleton and ran human-subject experiments [4].

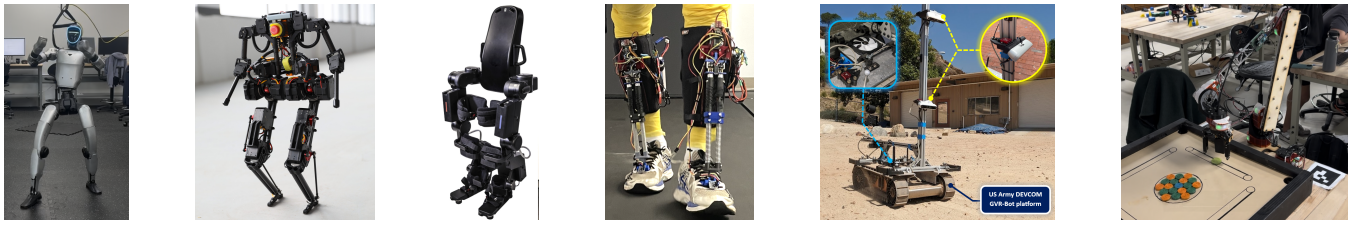
## PROFESSIONAL EXPERIENCE

**MIT Lincoln Laboratory** — Lexington, Massachusetts
*Research Intern* — JUN 2022 — SEP 2022

- Formulated additions to linear time-invariant control methodology for sensorless control of gimbal motor.
- Refactored Simulink and Jupyter API for experimentation of sensorless control algorithms on gimbal testbed.
- Achieved accurate sensorless estimation of steady-state electrical phase in under 20 seconds.

**AMBER Lab, Caltech** — Pasadena, California
*Research Intern* — JUN 2022 — SEP 2022
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## ROBOT DEPLOYMENT EXPERIENCE




Unitree G1, Westwood Robotics BRUCE, Wandercraft Atalante, Ankle Exoskeleton (*Axo*), DEVCOM GVR-BOT, 6-DOF Robot Arm

## TECHNICAL SKILLS

- **Programming Languages:** `Python, C/C++`
- **Software:** JAX, PyTorch, OpenCV, CVXPY, MuJoCo, Gymnasium, MjLab, Issac Lab, ROS1&2, (Ignition) Gazebo, Linux, Git, CMake, MATLAB/Simulink, Mathematica, Solidworks, Weights and Biases, SimpleFOC

## PUBLICATIONS

**Journal Publications:**

[1] **Janwani N.**, Novoseller E., Lawhern V., Tucker M. (2026). "MO-Playground: Massively Parallelized Multi-Objective Reinforcement Learning for Robotics." *IEEE Robotics and Automation Letters, 2026* [Webpage]

**Conference Publications:**

[2] **Janwani N.**†, Madabushi V†., Tucker M. "NaviGait: Navigating Dynamically Feasible Gait Libraries using Deep Reinforcement Learning." *IEEE International Conference on Robotics and Automation (ICRA), 2026* [Webpage]

[3] **Janwani, N.**†, Daş, E†., Touma, T., Wei, S. X., Molnar, T. G., & Burdick, J. W. "A Learning-Based Framework for Safe Human-Robot Collaboration with Multiple Backup Control Barrier Functions." *IEEE International Conference on Robotics and Automation (ICRA), 2024* [Paper] [Video]

**Patents**:

[4] Shaikewitz L., Tucker M., **Janwani N.**, Ames A., "Front-leg assistive exoskeleton" *US20240245569A1* [Paper]

## TEACHING EXPERIENCE

**Single Board Computers in Research, Caltech CS 12 Course** Pasadena, California
*Course founder and instructor* JAN 2023 — MAR 2024

- Developed 10-week course on prototyping robotic and computing systems for research applications.
- Taught 50 students in concurrent software and circuit design, sensor interfacing, microcontroller communication.
- Wrote hands-on labs in basic circuitry, hardware input, motor control and coached groups through final projects.
- Hired and trained an undergraduate teaching assistant to assist with course activities and grading.
- Granted $3000 from Caltech in funding for larger iteration of the course for the 2024 cycle

**Caltech Robotics and Computer Science** Pasadena, California
*Teaching Assistant* JAN 2022 — JUN 2024

- Designed a mobile robot as Robotics Head Teaching Assistant for a 35 student diversity, equity and inclusion program.
- Held office hours and wrote exam materials for classes of 100+ students in Experimental Robotics (ROS2: Python), Kinematics (ROS2: Python), Data Structures (Java), Software Design (C), and Computing Systems (x86 Assembly).

## AWARDS

**NSF Graduate Research Fellowship** APR 2024
Fellowship awarded by the National Science Foundation (NSF). Atlanta, Georgia

**IRIM Graduate Research Fellowship** APR 2024
Fellowship awarded by Georgia Tech's Institute of Robotics and Intelligent Machines (IRIM). Atlanta, Georgia

**Thomas A. Tisch Prize for Undergraduate Teaching in Computing and Mathematical Sciences** APR 2024
Awarded for development and instruction of new robotics course at Caltech. Pasadena, California

**Kiyo and Eiko Tomiyasu Research Fellowship** JUL 2023
Awarded a fully funded summer undergraduate research at the Burdick Group at Caltech. Pasadena, California

**Housner Fund** DEC 2023
Granted $3300 to develop a Caltech robotics course, supporting increased student interest. Pasadena, California

## SELECTED PROJECTS

- **Socially-Reasonable Humanoid Gestures:** collected dataset of human social gestures and retargeted to Unitree G1 humanoid robot. Utilized an LLM as a high-level social task planner to sequence retargeted motions.
- **Learning Humanoid Motion Priors Online:** leveraged VLMs to generate high-level task plans from natural language goals, and efficiently request one-shot demonstrations when the robot's skill library is insufficient.
- **Human-Robot Manipulation:** 5-DOF manipulator to play *Carrom* in a dynamic environment with human opponents.
- **Wikipedia Agent**: Sub-agents for retrieval augmented generation. Improved answer accuracy and speed over baselines.
- **AI Diagnostician:** Trained large vision models to predict pathologies from "CheXpert" chest X-rays.
- **Exploration-Based Planner:** Information-based exploration of an unknown space with probabilistic sensors.
- **Graph Search for Replanning:** Compared efficiency of D* from A* and Dijkstra for replanning on large graphs.
- **Quadruped Kinematics:** Wrote forward/inverse kinematics to simulate a quadrupedal robot designed in Solidworks.
- **Hierarchical Quadruped Planner:** Wrote hierarchical A* planner for quadrupedal locomotion on diverse terrains.
- **Mobile Robotics:** Implemented localization, obstacle avoidance, and odometry algorithms on hardware.